

Ἀρχιμήδης ὁ Συρακούσιος

Mathematics and Method in Archimedes of Syracuse

eul_aid: ixo

Period: Ἑλληνιστική [Hellenistic](#)
Dialect: Δωρική [Doric](#)
Domain: Μαθηματικά [Mathematics](#)
Format: Πραγματεία [Treatise](#)

Archimedes was a Greek mathematician, physicist, engineer, inventor, and astronomer from Syracuse, Sicily, born around 287 BCE. He likely studied in Alexandria and maintained correspondence with its scholars, but spent most of his life in Syracuse serving King Hieron II. He was killed by a Roman soldier during the siege of Syracuse in 212 or 211 BCE, despite orders for his protection, while reportedly engrossed in a mathematical diagram.

His surviving treatises, foundational to mathematics and physics, include *On the Sphere and Cylinder*, *Measurement of a Circle*, *On Spirals*, *On the Equilibrium of Planes*, *The Sand Reckoner*, *On Floating Bodies*, and *The Method of Mechanical Theorems*, which was rediscovered in 1906. Other works, like the *Stomachion* and the *Book of Lemmas*, survive in fragmentary or compiled form, while several, including *On Sphere-Making*, are lost.

Widely considered antiquity's greatest mathematician and scientist, Archimedes laid crucial foundations for integral calculus using the method of exhaustion. He established fundamental principles of mechanics and hydrostatics, such as the law of the lever and Archimedes' principle. His recovered works profoundly influenced the development of early modern science.

Works (14)

- Βιβλίον Ὑποθέσεων · [Book of Assumptions](#) · 30 passages
- Βοῦς Πρόβλημα · [Cattle Problem \(Long Version\)](#) · 4 passages
- Βοῦς Πρόβλημα · [Cattle Problem \(Short Version\)](#) · 2 passages
- Κύκλου Μέτρησις · [Measurement of the Circle](#) · 7 passages
- Μέθοδος πρὸς Ἑρατοσθένην · [Method to Eratosthenes](#) · 46 passages
- Ἀποσπάσματα · [On Archimedean Polyhedra](#) · 24 passages
- Περὶ Κωνοειδῶν καὶ Σφαιροειδῶν · [On Conoids and Spheroids](#) · 100 passages
- Περὶ Τῶν Ἐπιπλέοντων Σωμάτων · [On Floating Bodies](#) · 48 passages
- Περὶ Ἑλίκων Γραμμῶν · [On Spiral Lines](#) · 67 passages
- Περὶ Ἰσορροπίας Ἐπιπέδων · [On the Equilibrium of Planes](#) · 45 passages
- Περὶ Σφαίρας καὶ Κυλίνδρου · [On the Sphere and Cylinder](#) · 124 passages
- Παραβολῆς Τετραγωνισμός · [Quadrature of the Parabola](#) · 31 passages
- Ψαμμίτης · [Sand Reckoner](#) · 23 passages
- Στομάχιον · [Stomachion](#) · 3 passages